How Much of a Time Series Survives Its Image? A Cross-Domain Comparative Analysis of the Reconstruction Capacity of Image-Based Representations

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Abstract

Image-based representations have been widely studied for time-series classification, while recent work has also demonstrated their use for image-based synthesis and imputation. Yet successful generation does not establish what different encodings preserve or whether their images can be reliably decoded into signals. We investigate this question for binary, unsigned and signed recurrence plots (RP), Gramian angular summation and difference fields (GASF/GADF), and Markov transition fields (MTF), using the invertible-by-design XIRP as a reference. Alongside established inverses, we introduce an image-only geometric decoder for signed RP, a restricted-range GADF decoder, and an approximate MTF reconstruction procedure.

Across seven datasets from six domains, we evaluate temporal and spectral fidelity, robustness, computational cost and downstream classification utility, distinguishing encoding losses from 8-bit storage effects. Under stated conditions, signed RP, GASF, restricted-range GADF and XIRP allow numerically exact reconstruction from floating-point matrices; storage introduces addi-

tional error, particularly for ill-conditioned RP states. Unsigned RP retains a global-reflection ambiguity, whereas binary RP and MTF are intrinsically many-to-one.

These findings establish a reproducible comparison of reconstruction capabilities and limitations, informing future research on generating time series through image-domain models.
Keywords: Time-series imaging, Reconstruction, Recurrence plots, Gramian angular fields, Markov transition fields, Invertibility, Comparative analysis

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1. Introduction

Image-based representations transform time series into structured matrices that can be processed using techniques originally developed for images. Recurrence plots (RP), Gramian angular fields (GAF) and Markov transition fields (MTF) have been widely explored for time-series classification and related tasks [1, 2, 5, 6]. In classification, a representation need only preserve information relevant to the target task; reconstructing the original waveform is not necessarily required. However, applications that aim to recover or synthesise temporal signals introduce a different requirement: the encoded images must retain sufficient information to reconstruct meaningful time series.

Recent research has demonstrated the feasibility of exploiting image-based generative models for temporal data. Huang et al. [3] used Gramian angular representations with generative adversarial networks for traffic-data imputation. More recently, Hellermann and Lessmann [4] introduced the extended intertemporal return plot (XIRP), an invertible-by-design represen-

tation for time-series generation through image-based GANs. Their results demonstrated competitive predictive utility against specialised generative approaches and highlighted the importance of the inversion procedure. These developments establish image-based time-series generation as a promising research direction. Nevertheless, successful synthesis using particular representations does not establish the reconstruction capabilities of other encodings, nor the conditions under which their information can be recovered.

Understanding these properties requires distinguishing three aspects: the information intrinsically preserved by an encoding, the ability of a decoder to recover that information, and the stability of reconstruction after image storage or perturbation. This distinction is particularly relevant for representations whose inversion is ambiguous or has received comparatively limited attention. Moreover, accurately reconstructing an image produced by a known encoder does not guarantee that an arbitrary image generated by a learned model represents a valid time series. Characterising reconstruction is therefore a necessary preliminary investigation, rather than a demonstration of generative performance.

In this work, we establish a common framework for evaluating the reconstruction capabilities of image-based representations built from relationships between temporal samples or states. We examine binary, continuous unsigned and signed RP; Gramian angular summation and difference fields (GASF/GADF); and MTF. XIRP is included as an invertible-by-design reference developed for generative applications. Time-frequency and time-scale representations, such as spectrograms and scalograms, are outside the scope of this study.

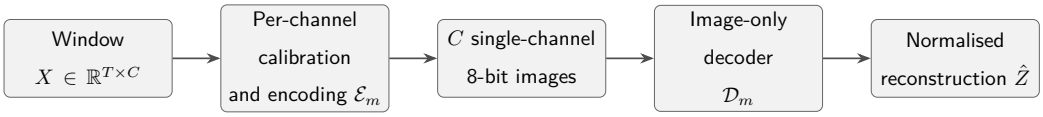


Figure 1: Controlled reconstruction protocol. Each channel is calibrated, encoded and stored as an 8-bit image. The decoder recovers a normalised trajectory from the image alone; the channel's minimum and maximum in the window are used only to return to physical units. Decoding before quantisation isolates reconstruction error from storage error.

We combine established reconstruction methods with new image-only procedures: a geometric decoder for signed RP, an anchored decoder for GADF under restricted angular normalisation, and an approximate representative-trajectory decoder for MTF. For recurrence plots, we distinguish the contribution of the encoding from that of the reconstruction algorithm through comparisons with the approaches of Hirata et al. [12] and Sipers et al. [14]. All methods are evaluated under a common 8-bit image-storage protocol, reconstructing normalised signals from the stored images alone; the minimum and maximum of each channel in the window are used only to recover physical units (Figure 1).

We address four research questions:

- *RQ1 — Recoverability.* What information can be recovered from each representation, in both the time and frequency domains, and how much is lost through encoding and 8-bit storage?

- *RQ2 — Reconstruction methods.* Under which conditions are the proposed decoders exact, and how do the encoding and decoding stages separately influence recurrence-plot reconstruction?

- *RQ3 — Robustness and stability.* How do image perturbations, quantisation and signal characteristics, including sampling rate and state-space conditioning, affect reconstruction fidelity?

- *RQ4 — Computational and downstream utility.* What are the computational costs of reconstruction, and to what extent do reconstructed signals preserve the performance of classifiers trained on original data?

The main contributions of this study are:

- *A reproducible reconstruction benchmark.* We systematically evaluate temporal image representations across seven public datasets from six domains, using a common protocol covering temporal and spectral fidelity, robustness, computational cost and downstream classification utility. This allows intrinsic representation losses to be distinguished from storage-related errors.

- *New reconstruction procedures and theoretical characterisation.* Our main methodological contribution is a geometric image-only decoder for signed RP, together with sufficient conditions for exact recovery. It is complemented by an anchored decoder for restricted-range GADF and an approximate reconstruction procedure for MTF, the latter as a control for an encoding that loses information. Controlled RP comparisons further separate the effects of directional information, distance preservation and decoder design.

- *An empirical foundation for image-based time-series generation.* We identify which representations admit reliable reconstruction, the ambi-

guities and limitations they retain, and the conditions affecting reconstruction stability. These findings provide a methodological basis for future generative studies, without assuming that reconstruction fidelity alone determines the quality of synthetic time series.

2. Related work

Temporal image encodings.. Recurrence plots (RP) visualise recurrences in an embedded state space, conventionally through a binary neighbourhood relation [5, 6]. Continuous RP variants preserve state distances; the modified signed RP of Lu and Tong additionally provides directional information [11]. Wang and Oates introduced Gramian angular summation and difference fields (GASF/GADF) and Markov transition fields (MTF) as image representations for classification and imputation [1, 2]. Their subsequent use in activity recognition and anomaly detection illustrates the value of these encodings as predictive features [7, 8]. Predictive usefulness, however, does not establish whether the original waveform can be recovered.

Reconstruction from encoded images.. The inverse problem has been investigated most directly for RP. Hirata et al. reconstruct state-space geometry from binary recurrence information, using graph-based distances and multidimensional scaling [12, 23]. For unthresholded RP, Sipers et al. establish identifiability conditions up to mean and global sign and develop a spectral reconstruction method, including experiments on disturbed and multilevel images [13, 14]. These results concern unsigned recurrence information and therefore do not resolve orientation using the directional content of a signed RP. For GASF, the diagonal inverse on a nonnegative normalisation is already

known and has supported image-based imputation [1, 9]. By contrast, GADF retains angular differences but not an immediately available absolute reference, and MTF quantisation discards within-state amplitudes.

Image-domain generation and the remaining question.. Recent work has extended temporal imaging beyond prediction: GAN-based image generation has been used for traffic-data imputation [3], while Hellermann and Lessmann introduced XIRP, a deliberately invertible encoding, for synthetic time-series generation with image-based GANs [4]. XIRP provides a useful reference because its diagonal explicitly stores the normalised trajectory. These contributions motivate a controlled study of what other images determine, how decoders use that information, and how 8-bit storage and perturbations affect reconstruction. For RP, we also separate encoding effects from decoder effects.

3. Mathematical formulation

3.1. Reconstruction problem and information retained

Let $x = (x_1, \ldots, x_T)$ denote a non-constant window of one signal channel. For representation m, the calibrated series $z_m = \mathcal{N}_m(x; x_{\min}, x_{\max})$ lies in a predefined normalised domain, specific to each representation: $[0, 1]$ for GASF, $[-0.7, 0.7]$ for GADF and $[1, 2]$ for XIRP, while MTF uses quantile states. The continuous RP images are unchanged by any increasing affine map of the series, because distances do not depend on a translation and the storage rule removes the scale (Section 4.1); so is the binary RP, because its threshold is set in each window by a fixed recurrence rate rather than in physical units; we therefore compute them on the window as recorded, which gives the same stored image as $z_m = (x - x_{\min})/(x_{\max} - x_{\min})$. The RP decoders recover the

trajectory up to a translation and a positive scale, which they fix by mapping its range onto $[0, 1]$; this is $\hat{z}_m$. Only then does $\mathcal{N}_m^{-1}$ use the calibration, and the implementation composes these two affine maps into one. It is encoded as a matrix, stored as an 8-bit image, and decoded according to

$$I_m = \mathcal{Q}_m\Big(\mathcal{E}_m(z_m)\Big), \qquad \hat{z}_m = \mathcal{D}_m(I_m), \qquad \hat{x} = \mathcal{N}_m^{-1}(\hat{z}_m; x_{\min}, x_{\max}).$$

Here $\mathcal{Q}_m$ is a predefined storage *rule*, possibly with a per-image affine range. The term *image-only* refers to $\hat{z}_m$: decoding never accesses the original window or window-specific side information. The pair $(x_{\min}, x_{\max})$ is the minimum and maximum of the channel in that window. It is used only to express $\hat{x}$ in physical units and never to decode $\hat{z}_m$; a future image-based generator would have to supply it separately (Figure 1).

For readability, encoding formulas below use x to denote the calibrated signal z_m. Three properties must be distinguished. *Identifiability* concerns whether two admissible normalised signals can produce the same encoded matrix, possibly up to a specified symmetry. *Decoder exactness* concerns whether $\mathcal{D}_m \circ \mathcal{E}_m$ recovers the original normalised signal under stated hypotheses, before storage. *Stability* concerns how errors in the image, including quantisation, propagate to the decoded signal. A many-to-one encoder cannot admit a universally exact inverse, irrespective of decoder design. Conversely, a mathematically identifiable signal may still be reconstructed inaccurately from an 8-bit image if the inverse is poorly conditioned.

3.2. Recurrence plots

For all RP variants we use adjacent two-sample states,

$$s_i = (x_i, x_{i+1}) \in \mathbb{R}^2, \qquad i = 1, \ldots, T-1,$$

that is, a two-dimensional delay embedding with unit delay [10]. The variants differ in how much information they retain about pairwise state displacements.

The *binary RP* records neighbourhood membership, $B_{ij} = \mathbb{1}[\|s_i - s_j\|_2 < \varepsilon]$, where we choose ε in each window so that a fixed share of the off-diagonal pairs, the recurrence rate, is recurrent. Thresholding discards the underlying distances, making the encoding many-to-one [5, 6]. The *continuous unsigned RP* stores $D_{ij} = \|s_i - s_j\|_2$, scaled by $\max_{ij} D_{ij}$. As a Euclidean distance matrix, D determines the embedded states up to translation and an orthogonal transformation. In particular, reflecting all signal values leaves D unchanged. Recovery up to global reflection therefore requires additional non-degeneracy assumptions; the unsigned image cannot determine the absolute orientation by itself [13].

The *signed RP* of Lu and Tong [11] associates a directional label with each distance. With $v = (1, 1)/\sqrt{2}$ and $\hat{u}_{ij} = (s_i - s_j)/\|s_i - s_j\|_2$ for nonzero displacements, its encoding is

$$R_{ij} = \sigma_{ij}\|s_i - s_j\|_2, \qquad \sigma_{ij} = \begin{cases} +1, & \|s_i - s_j\|_2 = 0 \text{ or } \hat{u}_{ij} \cdot v \geq \cos(\pi/4), \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

Its magnitude recovers the unsigned matrix, $|R| = D$. Directional information can additionally distinguish a signal from its global reflection: a pair is *sign-informative* when its displacement lies within $\pi/4$ of v or $-v$, in which case $\sigma_{ij} = -\sigma_{ji}$. Other nonzero pairs receive the same negative sign in both directions. Thus the signed RP adds information that is absent from the distance matrix, but this alone does not guarantee unique reconstruction. Its per-window affine image storage retains the quantised zero level through the

diagonal (Section 4.1).

For exact-input recovery, three issues must be resolved: the embedded geometry must be recoverable from distances; consecutive embedded states must determine a consistent one-dimensional trajectory; and the signed information must distinguish the remaining reflection. Proposition 1 states sufficient conditions for each, for the geometric decoder of Section 4.2. Let $\bar{s}$ be the mean state and $\sigma_1 \geq \sigma_2 \geq 0$ the singular values of the centred state matrix $S - \mathbf{1}\bar{s}^\top$, whose rows are $s_i - \bar{s}$.

Proposition 1 (Signed RP on exact input)**.** *Assume that*

(a) $\sigma_2 > 0$*, i.e. the centred states span* $\mathbb{R}^2$*;*

(b) there are no constants κ *and* $(\alpha, \beta) \neq (0, 0)$ *such that either* (R) $x_i + x_{i+2} = \kappa$ *for all* $i = 1, \ldots, T-2$*, or* (F) $\alpha(x_{i+2} - x_i) = \beta x_{i+1} + \kappa$ *for all* $i = 1, \ldots, T-2$*;*

(c) at least one pair of states is sign-informative.

Then the decoder of Section 4.2, applied to the floating-point signed RP, returns x *on the normalised domain. Without (c),* x *and* $-x$ *have the same signed RP, so no image-only decoder can tell them apart. Under (a), condition (b) holds if and only if one of the matrices* H_η *of step 3 has a simple zero eigenvalue and the other is positive definite, so it can be checked from the image; it holds, in particular, when the centred triples* (x_i, x_{i+1}, x_{i+2})*,* $i = 1, \ldots, T-2$*, have rank three.*

Proof sketch. Under (a), classical MDS recovers the states up to a translation and an orthogonal transformation [15]. Among those configurations, the overlap residual $b_i - a_{i+1}$ is constant only for the identity and its negative

when (b) holds, i.e. for the states of x and $-x$. Reflecting the series changes the sign of exactly the sign-informative entries, so the two signed RPs differ if and only if (c) holds, and the sign comparison of step 4 then selects x. The full proof is given in Supplementary Section S5.

None of these conditions follows from the others. For the sinusoid at a quarter of the sampling rate, $x_i = \sin(\pi i/2 + \phi)$, the states span the plane but (R) holds, every rotation satisfies the overlap, and the phases $\phi = 0.2$ and $\phi = 0.25$ give the same signed RP, so identifiability is lost altogether, not only up to sign. The series $x = (1, -1.05, 1.10, -1.15, 1.20, -1.25, 1.30, -1.35)$ satisfies (a) and (b), but every displacement lies close to the anti-diagonal, so no pair is sign-informative and x and $-x$ have the same signed RP. The conditions are sufficient, not necessary. In the benchmark, we checked (a)–(c) directly on all 61 079 channel series, with a relative tolerance of 10^{-9}: none fails, the smallest margins are $\sigma_2/\sigma_1 = 5.7e-06$ for (a) and 3.1e-06 for the (F) relation, and every series has a sign-informative pair (minimum share 6%). Consistently, the floating-point decoder recovered every one of them to numerical precision (Section 6.1).

The proposition concerns exact input, not stability after storage. For exact input, the doubly centred Gram matrix used by classical MDS has nonzero eigenvalues σ_1^2 and σ_2^2, so the gap separating the recovered two-dimensional subspace from the remaining eigenvalues is σ_2^2. By the Davis–Kahan theorem, a perturbation of that matrix rotates the subspace by an amount bounded by the perturbation divided by this gap [24]. We therefore report the scale-free ratio σ_2/σ_1 as an empirical indicator of geometric conditioning, not as a bound on the reconstruction error: quantisation, the recovery of the zero level

from the diagonal, the overlap orientation and the sign decision introduce further errors.

3.3. Gramian angular fields

GASF and GADF encode pairwise angular relations [1]. A normalised sample $\tilde{x}_i \in [-1, 1]$ is mapped to $\phi_i = \arccos \tilde{x}_i$, giving

$$G^{\mathrm{S}}_{ij} = \cos(\phi_i + \phi_j), \qquad G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j).$$

Their inverse properties differ substantially. For *GASF*, the diagonal is $G^{\mathrm{S}}_{ii} = \cos(2\phi_i)$. It is ambiguous over $[-1, 1]$ but injective when samples are normalised to $[0, 1]$, as in our experiments; there $\tilde{x}_i = \cos\!\left(\frac{1}{2} \arccos G^{\mathrm{S}}_{ii}\right)$. Thus GASF has a direct image-only inverse under the stated normalisation.

For *GADF*, the diagonal is identically zero and only relative angular differences are available. Recovering the series requires both a unique branch of arcsin and an angular reference. With per-window normalisation to $[-\alpha, \alpha]$, the maximum angle separation is

$$\max_{i,j} |\phi_i - \phi_j| = \pi - 2 \arccos \alpha < \frac{\pi}{2} \qquad \text{if} \quad \alpha < \frac{1}{\sqrt{2}}.$$

We use $\alpha = 0.7$, so angular differences remain on the principal branch. Because every non-constant calibrated window reaches both endpoints, the global offset can then be fixed using the image itself (Section 4.4). This conditional inverse does not extend to arbitrary GADF normalisations: outside the restricted interval, branch ambiguities may remain.

3.4. Markov transition fields

MTF first assigns each sample to a quantile state $q_t \in \{1, \ldots, Q\}$ and estimates the transition probabilities $W_{pq} = P(q_{t+1} = q \mid q_t = p)$ [1]. The

image is then defined by $M_{ab} = W_{q_a q_b}$. It retains information about the temporal arrangement of discrete states but discards within-state amplitudes; moreover, numerical state ordering cannot generally be inferred uniquely from the image. Different waveforms can therefore yield the same MTF. In this setting reconstruction is necessarily approximate: the decoder selects a representative trajectory rather than recovering a uniquely determined original.

3.5. Extended intertemporal return plot

XIRP was proposed specifically for image-based time-series generation [4]. For a strictly positive normalised series it combines intertemporal log returns with an explicit record of the sample values:

$$R_{ij} = \begin{cases} \log(x_i/x_j), & i \neq j, \\ x_i, & i = j. \end{cases}$$

The diagonal thus provides an exact inverse before storage, while every off-diagonal column supplies additional estimates through $x_i = x_j \exp(R_{ij})$. We map each channel to $[1, 2]$ with the common calibration and include XIRP as an *invertible-by-design reference*, not as another conventional classification encoding.

Table 1 summarises these recoverability properties; Figure 2 displays the actual stored images for representative signal windows.

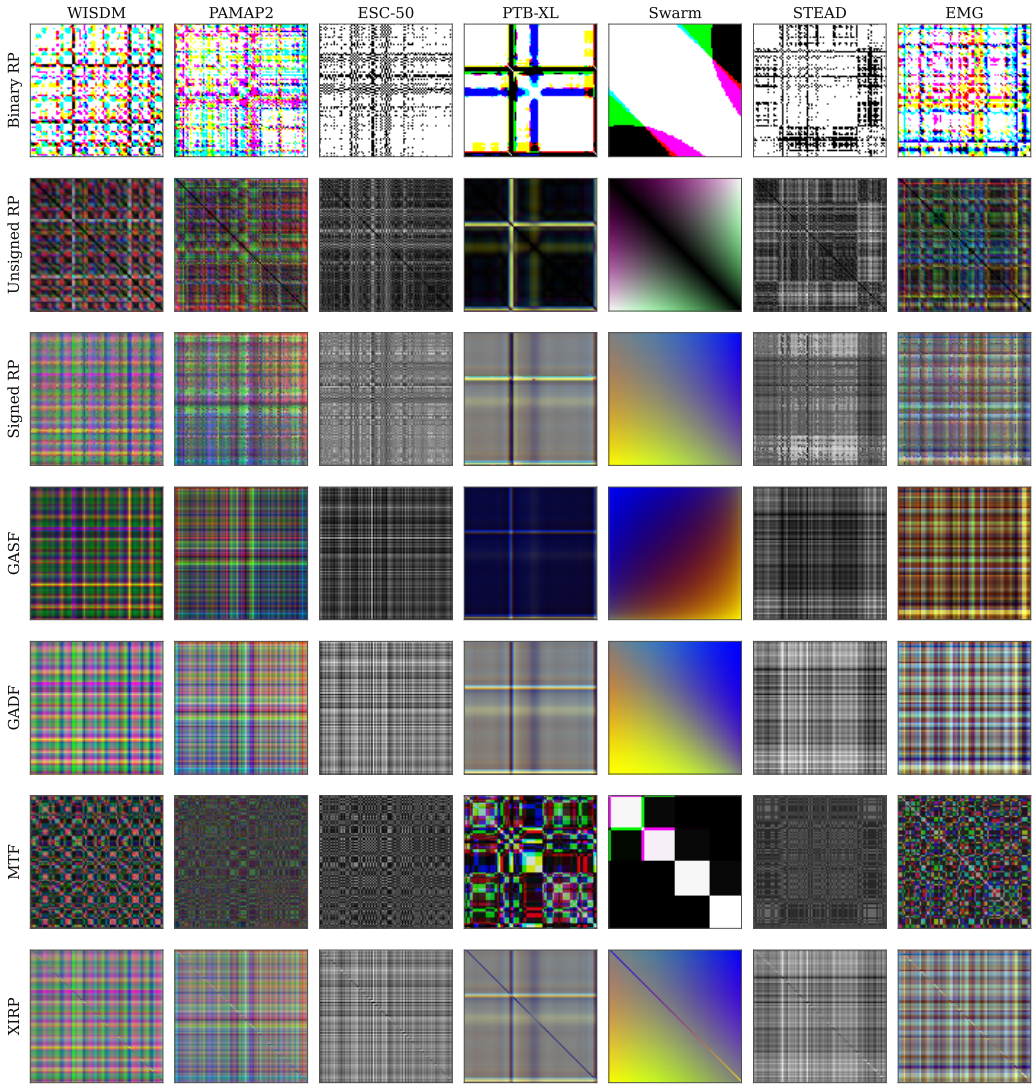


Figure 2: Representative 8-bit encoding images ($T = 128$; one window per dataset). Each channel is stored as its own single-channel image; for display, the first three channels are shown as the red, green and blue planes of one picture, and the single-channel ESC-50 appears in grey. The binary RP uses a recurrence rate of 0.75. The red (first) channel is decoded in Figure 3.

Table 1: Information retained by each encoding and reconstruction approaches evaluated. Exactness refers to unquantised matrices under the stated conditions; decoders operate on normalised signals using the image alone.

Encoding	Image determines	Exact recovery	Decoders (origin)
Binary RP	Recurrence neighbourhoods	No (many-to-one)	Hirata's graph method, with our sample reconstruction
Unsigned RP	State geometry	Up to reflection, under (a)–(b) of Prop. 1	Hirata after thresholding [12]; Sipers on circular RP [14]; geometric method without sign (this work)
Signed RP	Geometry and directional information	Conditional (Prop. 1)	Hirata and Sipers with adapted sign selection; proposed geometric decoder
GASF	Angular sums	Yes on $[0, 1]$	Diagonal inverse [1]
GADF	Angular differences	Yes when $\alpha < 1/\sqrt{2}$ with bounded normalisation	Restricted-branch and boundary-anchor decoder (this work)
MTF	Discrete-state transitions	No (many-to-one)	Spectrally ordered representative (this work)
XIRP	Returns and explicit diagonal values	Yes	Inversion by mean [4]

4. Decoders

The previous section describes what each encoding retains; here we describe how that information is used to reconstruct a trajectory. All methods operate on the stored image. Published methods are distinguished from our adaptations, and the RP experiments compare both changes of encoded information and changes of decoder.

4.1. Storage and image-only decoding

An encoded matrix is converted to an 8-bit image through the storage rule $\mathcal{Q}_m$. GASF and GADF use the known range $[-1, 1]$, MTF uses $[0, 1]$, and XIRP uses $[1, 2]$ on the diagonal and $[-\log 2, \log 2]$ off it. For signed RP, the per-image affine range is not supplied to the decoder. Instead, the quantised diagonal identifies the zero level; subtracting its median recovers a signed matrix up to an unknown positive factor. The subsequent geometric steps and final range normalisation are invariant to this factor. All choices needed to decode the *normalised* trajectory are therefore made from the stored image and globally fixed method settings, not from the original window.

4.2. Geometric reconstruction of signed RP

The proposed decoder recovers state geometry from distances, finds the orientation compatible with consecutive overlapping states, and finally uses directional signs to resolve the remaining reflection. For a length-T signal, the procedure is:

1. *Distance recovery.* Re-centre the stored pixel matrix at its diagonal zero level and symmetrise the absolute values:

$$R^{\mathrm{obs}} = \mathrm{PNG} - \mathrm{median}_i \, \mathrm{PNG}_{ii}, \qquad D_{ij} = \frac{|R^{\mathrm{obs}}_{ij}| + |R^{\mathrm{obs}}_{ji}|}{2}.$$

2. *State geometry.* Apply classical multidimensional scaling [15]. With $J = I - \mathbf{1}\mathbf{1}^\top/(T-1)$, form $B = -\frac{1}{2}JD^{\circ 2}J$. Its two leading eigenpairs give $Y = V_2\Lambda_2^{1/2}$, a state configuration determined up to an orthogonal transformation.

3. *Overlap-compatible orientation.* Adjacent states must share one sample: for oriented rows (a_i, b_i), ideally $b_i = a_{i+1}$. Parameterise the unknown orthogonal transform by $Q = \mathrm{rot}(\theta)\,\mathrm{diag}(1, \eta)$, $\eta \in \{-1, +1\}$. For each η, the residual has the form $b_i - a_{i+1} = A_i \cos\theta + B_i \sin\theta$, with

$$A_i = \eta Y_{i2} - Y_{i+1,1}, \qquad B_i = -\eta Y_{i1} - Y_{i+1,2}.$$

   After removing the mean residual, the least-squares problem reduces to $\min_{\|u\|=1} u^\top H_\eta u$, where $u = (\cos\theta, \sin\theta)$ and $H_\eta = \tilde{C}^\top\tilde{C}$, $\tilde{C} = [\tilde{A}\ \tilde{B}]$. The smallest-eigenvalue eigenvector of each 2×2 matrix, taken with both signs, gives four candidates.

4. *Directional disambiguation.* Re-encode each candidate using the signed RP and select the one whose unit-norm matrix is closest to the unit-norm observed matrix in Frobenius norm. This comparison uses the image itself, never the original signal.

5. *Sample assembly.* For the selected candidate, let $\delta^\star = -\,\mathrm{mean}_i(b_i - a_{i+1})$ and correct $a_i \mapsto a_i - \delta^\star/2$, $b_i \mapsto b_i + \delta^\star/2$. Set $\hat{x}_1 = a_1$, $\hat{x}_i = (b_{i-1} + a_i)/2$ for $2 \le i \le T-1$, and $\hat{x}_T = b_{T-1}$. Finally map its range to the method's normalised domain.

The orientation problem is linear in T, while a dense eigendecomposition dominates the implementation cost, $O(T^3)$.

4.3. RP baselines and controlled comparisons

Geometric decoder without sign.. To isolate the information contributed by directional labels, we also apply the geometric procedure to unsigned distances. We retain the MDS, overlap-fitting and sample-assembly steps, but cannot perform the signed-image comparison. On non-degenerate exact inputs, both orientations have the same distance image, so an image-only decoder cannot determine which one produced the original.

Hirata et al.. As a binary-RP reference, we implement the graph-distance and MDS construction of Hirata et al. [12, 23]. Thresholding gives a recurrence graph with edges $B_{ij} = 1$. For neighbour sets $G_i = \{k : B_{ik} = 1\}$, an edge receives weight $1 - |G_i \cap G_j|/|G_i \cup G_j|$. All-pairs shortest paths provide estimated distances, followed by MDS. Converting the resulting coordinates into a length-T series is our post-processing: retain the leading coordinate, append its mean, and rescale the range. We set the recurrence rate to 0.75 before benchmarking and report a sensitivity analysis over 0.02–0.75 (Section 6.2). When starting from a signed RP, the same unsigned graph determines the geometry; its directional signs are used only in an additional orientation-selection step. Thus this adaptation should not be confused with Hirata's original output procedure.

Sipers et al.. For continuous unsigned RP, we implement the spectral method of Sipers et al. [14]. It exploits the factorisation of two-dimensional Fourier coefficients of the squared distance matrix and reconstructs the signal up to a sign. The published construction assumes periodicity, so we provide it with a *circular* RP that includes the wraparound state (x_T, x_1). Image-derived frequency components below 10^{-4} of the strongest estimated power

are discarded. This frequency threshold, unrelated to Hirata's recurrence threshold, was set in an exploratory sweep (Supplementary Section S10) and confirmed on calibration windows that no evaluation uses, where it minimises Sipers' own 8-bit error: 100 windows per dataset at $T = 128$ outside both the fidelity and the classification samples, on the 6 datasets that have such windows (all EMG windows are evaluated). It is the best value whether the reflection is fixed with the original series or chosen from the signed image (Supplementary Section S11). Signed RP reconstruction adds our directional orientation-selection step. Because circularisation changes the encoded matrix, decoder comparisons are also performed on the *same circular image* (Supplementary Table S8), rather than attributing an encoding change to Sipers' method.

Pairing of the comparisons.. Every combination is evaluated on the same windows and channels, so all comparisons are paired by series. Hirata's reconstruction and the geometric decoder, with and without sign, read the same conventional RP image; Sipers' reconstruction reads the circular RP, a different image of the same series, so in the benchmark it is paired by series but not by image. Only Supplementary Table S8, where both decoders read the identical circular image, isolates the decoder from the encoding.

4.4. Restricted-range GADF decoder

GADF provides angular differences but no absolute angle. Our procedure reconstructs a reference angle from the image and then recovers the remaining angles from relative differences. The method assumes the fixed normalisation $[-\alpha, \alpha]$ with $\alpha = 0.7 < 1/\sqrt{2}$.

1. *Image-derived boundary anchor.* Compute row means

$$\mu_i = \frac{1}{T}\sum_j G^{\mathrm{D}}_{ij} = \rho \sin(\phi_i - \psi),$$

where $\rho \geq 0$ and ψ lies within the angular interval spanned by the samples. That interval is narrower than $\pi/2$, so the row means are monotone in ϕ_i. Thus $e = \arg\max_i |\mu_i|$ is an endpoint sample. Fix $\phi_e = \arccos(-\alpha)$ when $\mu_e > 0$ and $\phi_e = \arccos \alpha$ otherwise.

2. *Relative-angle recovery.* For pivot row p, recover every angle by

$$\hat{\phi}^{(p)}_j = \phi_e - \arcsin G^{\mathrm{D}}_{ep} - \arcsin G^{\mathrm{D}}_{pj}.$$

3. *Quantisation mitigation.* Average the angle estimates obtained from the five rows with the smallest $|\mu_p|$, then return $\hat{\hat{x}}_j = \cos\Big(\frac{1}{5}\sum_p \hat{\phi}^{(p)}_j\Big)$.

Under the normalisation assumptions the decoder is exact before storage and has $O(T^2)$ complexity. Pivot averaging is a numerical choice to mitigate pixel perturbations, not a condition for identifiability. The restricted normalisation is what makes the series identifiable; the procedure is what recovers it from the image alone. An elementary inversion of relative angles needs one absolute angle from outside the image, such as the stored value of one sample. Here the restricted range makes the row means monotone in the angle, so the image identifies an endpoint sample and its angle; and averaging over pivot rows spreads the quantisation error of any single row instead of propagating it to every sample. The contribution is therefore the image-derived anchor and the pivot averaging; the invertibility itself follows from the normalisation.

4.5. GASF, MTF and XIRP reference decoders

GASF.. We apply the established diagonal inverse on the $[0, 1]$ domain [1]. Its error after 8-bit storage can therefore be distinguished from the exact-input

inverse.

MTF.. Since the image does not identify within-state amplitudes, our decoder constructs a representative sequence rather than an exact inverse. The T image rows are grouped into $Q = 4$ clusters, a value fixed in advance (Supplementary Section S3); we estimate state-to-state transitions from the corresponding image blocks and order the clusters using the Fiedler vector of a symmetrised transition-graph Laplacian [16]. State k is assigned the bin midpoint $(k - \frac{1}{2})/Q$. Neither absolute state order nor within-state amplitudes are guaranteed by the image, so the resulting error combines irreversible encoding loss with approximation choices of the decoder.

XIRP.. We use the inversion-by-mean procedure of Hellermann and Lessmann [4]. Since the original normalised values appear on the matrix diagonal, this method serves as an invertible reference whose remaining error after quantisation can be measured directly.

5. Experimental protocol

5.1. Datasets and windows

Table 2 lists the seven public datasets: human activity (WISDM [17], PAMAP2 [25]), environmental audio (ESC-50 [26]), electrocardiography (PTB-XL [27]), geomagnetism (Swarm [28]), seismology (STEAD [29]) and surface electromyography (EMG [30]). Every channel is encoded, stored and decoded as its own single-channel image with its own calibration. Recordings are cut into non-overlapping windows of $T \in \{64, 128, 256\}$ samples that never cross a label change or a time gap. Blocks with fewer than 200 windows (EMG at

$T = 256$) are not reported. Preprocessing and sample sizes are detailed per dataset in Supplementary Section S1.

Table 2: Datasets. Windows of 128 samples available after segmentation; the group is the unit that never appears on both sides of a split.

Dataset	f_s (Hz)	Channels	Task	Windows	Group
WISDM	20	3	5 activities	8 172	subject (51)
PAMAP2	100	9	12 activities	15 085	subject (9)
ESC-50	8 000	1	50 sound classes	624 000	source recording
PTB-XL	100	12	5 classes, multilabel	21 000	patient
Swarm MAG	1	3	none	62 757	–
STEAD	100	3	earthquake vs noise	92 000	event / station
EMG gestures	100	8	6 gestures	880	subject (36)

5.2. Fidelity, robustness and cost

Fidelity uses 500 windows per dataset and length drawn with a fixed seed, and 3 000 for the single-channel ESC-50, so that every block holds between about 1 500 and 6 000 channel series. Writing $r = \max_i x_i - \min_i x_i$, it is measured per channel series by

$$\mathrm{NMAE} = \frac{1}{rT} \sum_i |x_i - \hat{x}_i|,$$

the Pearson correlation, the L_1 distance between unit-mass Welch spectra [18] and the relative energy error; constant series are excluded. Each decoder is run on the floating-point matrix and on the 8-bit image. Robustness adds zero-mean Gaussian noise of $\sigma \in \{2, 4, 8, 16\}$ intensity levels to every pixel of the stored image, which is then clipped and rounded, with two replicates per level on 200 windows per dataset and length. Cost is the single-thread time

to encode, store and decode and the peak traced memory, for T from 64 to 1 024. The sampling-rate study reads 3 000 PTB-XL records at 500 Hz and resamples them with an anti-aliased polyphase filter to 250, 100 and 50 Hz, with windows of 128 samples and, as a control, of a fixed duration of 1.28 s.

5.3. Classification utility

Classifiers are trained on original windows and evaluated on the original and on every reconstruction of the same held-out windows. Up to 6 000 windows per dataset and length are split into three grouped folds that keep every subject, patient, event, station or source recording on one side, stratified by class for single-label data; per-channel standardisation uses training statistics only, and reconstructions are produced independently per window, so no information crosses the split. KNN, an RBF SVM, a random forest, an LSTM, a GRU and a one-dimensional CNN are used with standard settings (Supplementary Section S4). Accuracy, or macro-F1 for the multilabel PTB-XL, is averaged over the three folds. Swarm has no labels. ESC-50 windows of 128 samples span 16 ms of sound, which leaves every classifier near chance, so audio is evaluated on the official ESC-10 subset at 2, 4 and 8 kHz with windows of 512 and 1 024 samples, silent windows removed and accuracy scored per clip.

5.4. Statistical protocol

All comparisons are paired over the same series; differences between decoders are summarised by medians of per-series differences, averaged across datasets with equal weight. A reconstruction is declared *equivalent* to the

original for a classifier when the 95% bootstrap interval of the accuracy difference, resampling whole groups [19, 20], lies within ± 0.01, which corresponds to two one-sided tests at the 2.5% level [22, 21]; a non-significant difference is not taken as evidence of equivalence. Equivalence is assessed on single-label data; for PTB-XL only the drop in macro-F1 is reported.

6. Results

6.1. Fidelity and storage (RQ1)

Table 3 gives the 8-bit error of every combination at $T = 128$; the other lengths behave alike (Supplementary Tables S4 and S5). Over the 20 reported dataset–length blocks, 61 079 channel series were reconstructed by every combination without decoder failure.

On the floating-point matrix, the geometric decoder on the signed RP, GASF, GADF and XIRP recover the series to machine precision: the median NMAE per block never exceeds 1.4e-13 for the signed RP and 5.6e-17, 8.4e-17 and 2.8e-16 for GASF, GADF and XIRP. Over every series of every block, the largest error is 1.6e-11 for the signed RP and 8.0e-13, 3.6e-16 and 6.3e-16 for GASF, GADF and XIRP. For these four, the 8-bit error is therefore storage error. It stays between 0.0009 and 0.0022 for the signed RP on six datasets and rises to 0.012–0.017 on Swarm; GASF and GADF are of the same order, and XIRP is about an order of magnitude lower. Sipers' reconstruction on the circular signed RP comes close, between 0.0028 and 0.0155. Without the frequency threshold it is exact in floating point but less accurate after storage; with the threshold used here its floating-point error reaches a median of 1.7e-03 per block (Supplementary Section S10).

The remaining combinations reach an NMAE between about 0.03 and 0.34, for two different reasons (Figure 3; spectral and correlational metrics in Supplementary Table S6). On the unsigned RP, the geometric and Sipers' decoders keep the spectrum as well as on the signed RP, because a reflection leaves the spectrum unchanged; its pointwise error and its correlation of about zero come from returning the reflection in about half of the series. Hirata's reconstruction and the MTF representative also distort the spectrum and the energy: they lose the waveform, not only its sign.

Table 3: Mean 8-bit NMAE of every encoding–decoder combination at $T = 128$ (500 windows per dataset, 3 000 for ESC-50; every channel). Uns./Sig.: unsigned and signed RP.

	Hirata		Sipers		Geometric (ours)					
Dataset	Uns.	Sig.	Uns.	Sig.	Uns.	Sig.	GASF	GADF	MTF	XIRP
WISDM	0.2411	0.1044	0.1954	0.0043	0.1946	0.0010	0.0015	0.0013	0.2838	0.0002
PAMAP2	0.2345	0.1059	0.1884	0.0050	0.1899	0.0015	0.0016	0.0013	0.2771	0.0002
ESC-50	0.2184	0.0840	0.1916	0.0059	0.1922	0.0017	0.0016	0.0011	0.2740	0.0002
PTB-XL	0.3313	0.1982	0.2680	0.0055	0.2874	0.0016	0.0029	0.0023	0.3404	0.0002
Swarm	0.2778	0.0283	0.2837	0.0045	0.2547	0.0124	0.0026	0.0015	0.2979	0.0002
STEAD	0.2044	0.0868	0.1802	0.0051	0.1771	0.0013	0.0015	0.0011	0.2668	0.0002
EMG	0.1979	0.1205	0.1669	0.0039	0.1680	0.0009	0.0013	0.0022	0.2834	0.0006

6.2. Recurrence plots: encoding versus decoder (RQ2)

Table 3 separates five effects. The spectrum is unchanged by a reflection of the series, so it shows what each decoder keeps of the waveform independently of its orientation. *Sign.* On the unsigned RP, the geometric decoder keeps the spectrum as well as on the signed RP (PSD-shape distance 0.0007–0.0151, against 0.0034–0.0187), while its correlation with the original stays

within ±0.05: it recovers the shape and loses only the orientation, which the sign restores from the image alone (Figure 3). *Thresholding.* Hirata's reconstruction from the binarised RP distorts the spectrum itself (0.06–0.61): once the image is binary, the waveform is lost, not only its sign. *Information and algorithm.* On the same signed RP, the geometric decoder is one to two orders of magnitude more accurate than Hirata's on six datasets; on Swarm the gap narrows to a factor of about two. This comparison changes both the information used, continuous distances against a thresholded graph, and the algorithm, two-dimensional MDS with overlap assembly against graph geodesics with our one-coordinate post-processing, so the gap cannot be attributed to the distances alone. *Decoder on continuous distances.* Sipers' reconstruction also uses the continuous distances; with the sign it is a median of 0.0015 per series above the geometric decoder, which on the same circular image is 1.8 to 4.8 times more accurate on every dataset (Supplementary Table S8). It is ahead in the benchmark only on Swarm, and the gain comes from the encoding: adding the wrap state raises the median σ_2/σ_1 from 0.0003 to 0.15 and lowers the geometric decoder's error from 0.0122 to 0.0025, below Sipers' 0.0046 on the same image, while on other datasets it changes by at most 0.0002. *Threshold choice.* These comparisons do not depend on the recurrence rate fixed for Hirata: choosing the best rate per dataset with knowledge of the result lowers its error by at most 0.025 on the unsigned and 0.010 on the signed RP, leaving it between 0.02 and 0.31 (Supplementary Table S7).

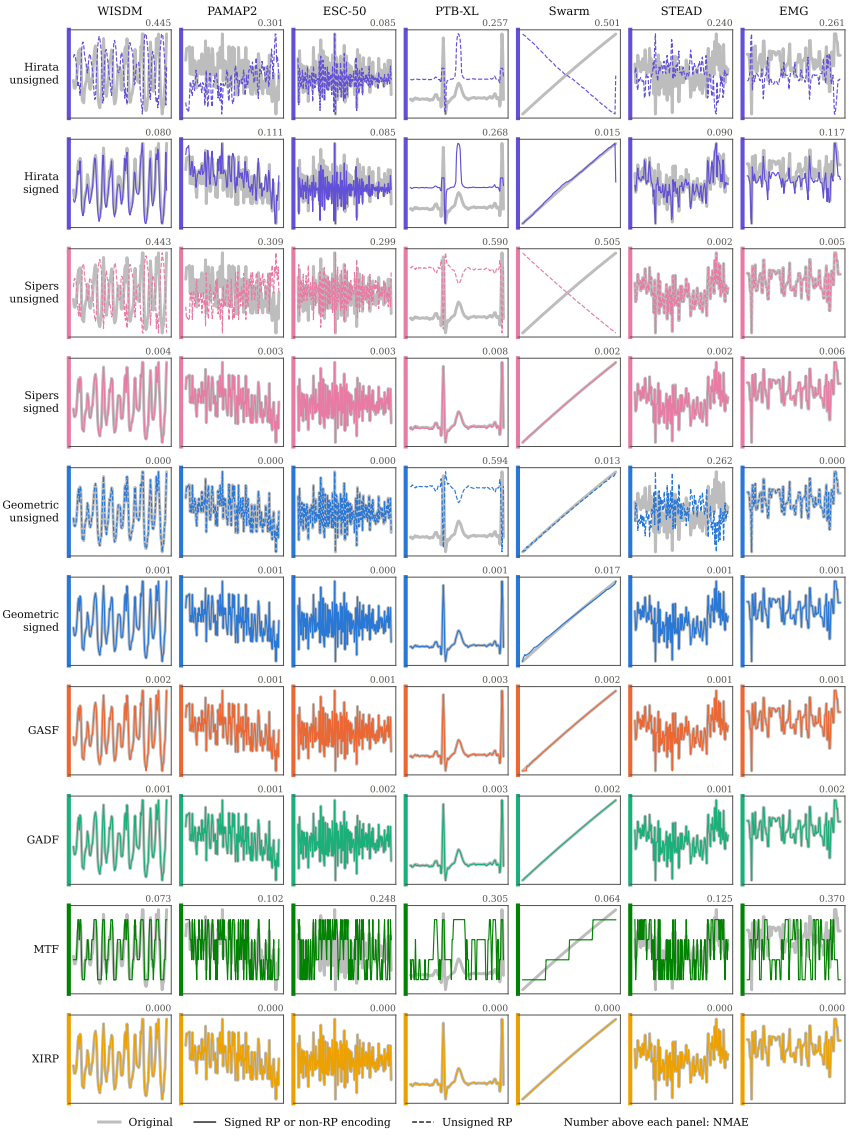


Figure 3: The first channel of the windows of Figure 2, decoded from its 8-bit image. Each row is one encoding–decoder combination, in the order of Table 3, and each column one dataset; the original is the grey band, the reconstruction is drawn on top of it, and the NMAE of the panel is given above it. On Swarm, whose states are nearly collinear, the signed RP deviates visibly at the start of the window.

6.3. Signal characteristics and stability (RQ3)

Storage error is expected to be amplified when the states are nearly collinear, because classical MDS then recovers a two-dimensional subspace separated by a small eigengap [24], and the data agree. None of the 54 040 series with $\sigma_2/\sigma_1 \geq 0.1$ exceeds an 8-bit NMAE of 0.01, whereas 85% of the series with $\sigma_2/\sigma_1 < 0.01$ do. The ratio is an empirical indicator, not a complete explanation: it separates the regimes at the two ends but does not determine which series in between fail. Nor does the share of sign-informative pairs: slow series are informative in nearly every pair, yet their states are the most collinear. Swarm, sampled at 1 Hz, is the limiting case and the only dataset where the signed RP is not in the low-error group.

The sampling rate changes how far apart consecutive samples are, and with it the conditioning (Figure 4b–c). On PTB-XL with windows of 128 samples, from 50 to 500 Hz the median σ_2/σ_1 falls from 0.61 to 0.08, the share of series below 0.1 grows from 0% to 67%, and the 8-bit NMAE of the signed RP rises from 0.0009 to 0.0076, while GASF and GADF stay between 0.0018 and 0.0032 and XIRP is unaffected. A fixed number of samples spans a shorter time at a higher rate, but the trend persists at a fixed duration of 1.28 s, from 0.0012 to 0.0079 (Supplementary Table S9). Sipers' reconstruction rises with the rate as well (Figure 4b), so the effect belongs to the RP, not to one decoder. The resampled and the distributed 100 Hz files give the same error (0.0015 and 0.0015). Resampling also changes the spectral content of each window, and the study covers one dataset, so the trend is attributable to the rate as realised by anti-aliased resampling of PTB-XL, not to the rate in isolation.

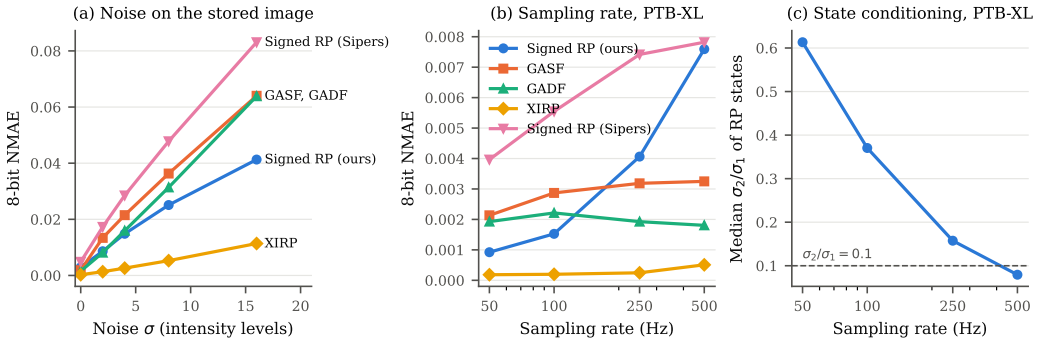


Figure 4: (a) Mean 8-bit NMAE against Gaussian noise added to the stored image ($T = 128$, mean over the seven datasets, two replicates per level). (b) 8-bit NMAE and (c) median conditioning σ_2/σ_1 of the RP states against the sampling rate (PTB-XL, $T = 128$, anti-aliased resampling from 500 Hz). Hirata's reconstruction, the geometric and Sipers' decoders on the unsigned RP and MTF stay between 0.08 and 0.34 throughout and are omitted.

6.4. Classification utility (RQ4)

Six classifiers, five labelled datasets and three lengths give 84 combinations on 72 917 windows (Table 4). On the original windows, accuracy ranges from 0.17 to 0.81 and PTB-XL macro-F1 from 0.14 to 0.61; the lowest values (KNN on EMG) are near chance, where drops are less informative. The four combinations that are exact in floating point never lose more than 0.007 and are equivalent within ±0.01 in nearly every single-label combination; where equivalence is not established, the observed difference is still below 0.01 and the interval is too wide rather than the drop large. Sipers' reconstruction on the circular signed RP joins this group, with a largest drop of 0.008, but is equivalent in fewer combinations (59 of 66, against 63 for the geometric decoder), consistent with its larger 8-bit error. Without the sign, Sipers' reconstruction, Hirata's on either encoding, the geometric decoder on the

unsigned RP and MTF lose between 0.08 and 0.12 on average and up to 0.40: whichever decoder is used, the unsigned RP loses the orientation and with it the accuracy. The same separation holds per clip on ESC-10 (18 000 windows, 36 combinations), where the exactly decodable combinations lose at most 0.017. Results per dataset and per ESC-10 block are in Supplementary Section S14.

Table 4: Drop from the original accuracy (macro-F1 on PTB-XL) for classifiers trained on original windows and tested on reconstructions. $|\Delta| \leq 0.01$: share of combinations whose observed drop is within 0.01. Equivalent: 95% group-bootstrap interval within ± 0.01, over the single-label combinations.

	Five datasets, $T \in \{64, 128, 256\}$				ESC-10, per clip	
Decoder	Mean drop	Max drop	$\lvert\Delta\rvert \leq 0.01$	Equivalent	Mean drop	Equivalent
Hirata / uns.	0.120	0.376	19%	11/66	0.071	0/36
Hirata / sig.	0.080	0.283	26%	12/66	0.060	0/36
Sipers / uns.	0.098	0.371	19%	13/66	0.021	0/36
Sipers / sig.	0.000	0.008	100%	59/66	0.001	28/36
Ours / uns.	0.099	0.359	21%	13/66	0.024	0/36
Ours / sig.	0.000	0.007	100%	63/66	0.001	33/36
GASF	0.000	0.005	100%	63/66	0.001	34/36
GADF	0.000	0.002	100%	63/66	0.000	32/36
MTF	0.099	0.400	19%	7/66	0.106	0/36
XIRP	0.000	0.002	100%	65/66	0.000	36/36

6.5. Robustness (RQ3) and cost (RQ4)

With noise of $\sigma = 16$ intensity levels on the stored image, the mean NMAE at $T = 128$ rises to 0.011 for XIRP, 0.041 for the signed RP, and 0.064

and 0.064 for GASF and GADF, and 0.083 for Sipers' reconstruction on the circular signed RP (Figure 4a). The geometric decoder on the signed RP degrades less than both angular fields on 6 of the 7 datasets, with the three level on Swarm, and less than Sipers' on 7 of the 7. The lossy combinations move by at most 0.020, because their error is already dominated by what is discarded before storage. Supplementary Table S10 gives every dataset.

Hirata's reconstruction has the steepest cost, with an exponent of about 2.6 due to all-pairs shortest paths (Supplementary Table S14). The geometric decoder on the signed RP is 2.9 times faster at $T = 256$ and 5.2 times faster at $T = 1024$, and costs about as much as Sipers' reconstruction at $T = 256$ and half as much at $T = 1024$. GASF, GADF, XIRP and MTF are one to two orders of magnitude cheaper. Memory grows quadratically for all of them.

7. Discussion

Invertibility is not stability. The signed RP with the geometric decoder, GASF, GADF on its restricted range and XIRP are exact on unquantised matrices under the conditions of Section 3, and every benchmark series met them when they were checked directly. After 8-bit storage they stay close to the original and keep the accuracy of classifiers trained on original signals, but exactness before quantisation does not determine how errors propagate afterwards. The signed RP is the clearest case: its storage error depends on the conditioning of the states, and it grew for slowly varying windows such as Swarm at 1 Hz and PTB-XL resampled to 500 Hz, for Sipers' decoder as for ours, while GASF and GADF stayed level. XIRP, whose column proposals are redundant, was the least sensitive to quantisation and to pixel noise; its lower error reflects

that it stores the series by design and says nothing about whether its images are better inputs for classification or generation. These observations hold for the storage and perturbations studied here, 8-bit images and independent Gaussian pixel noise; other formats or structured perturbations may behave differently.

What an encoding retains and how well it is decoded are different questions. The unsigned RP keeps the geometry of the states but not their orientation: both the geometric decoder and Sipers' recover the spectrum and lose the sign, so this loss belongs to the encoding, not to a decoder. The binary RP and MTF are many-to-one, so no decoder can be exact for every series; the evaluated decoders lose most of the waveform, and better approximations, for instance with learned priors, are not ruled out but cannot guarantee the encoded series either. Once the information is present, the decoder matters: on the same circular signed RP the geometric decoder is 1.8 to 4.8 times more accurate than Sipers' (Supplementary Table S8). The encoding can matter as much: on Swarm, Sipers' reconstruction is ahead in the benchmark because circularisation adds a wrap state that improves the conditioning, and on that same circular image the geometric decoder overtakes it. Comparisons between RP decoders should therefore fix the encoding before attributing a difference to the algorithm, and the comparison with Hirata's reconstruction, which changes both, should be read in that light.

From encoder outputs to generated images. Every image studied here is the output of a known encoder applied to a real series. Decoding such images does not show that a generator can produce images with the same structure, nor that the series decoded from generated images are good synthetic data.

A generated signed RP need not be a signed Euclidean distance matrix with a zero diagonal and consistent signs, a generated GADF need not stay on the restricted branch, and the per-window calibration $(x_{\min}, x_{\max})$ would have to be generated or fixed. Hellermann and Lessmann address the first problem by designing XIRP to be invertible [4]; our results show which other encodings can be decoded once their structure is respected, and under which conditions.

Limitations. Images are deterministic outputs of known encoders; pixel noise measures local stability only. Physical units need the window's own channel extremes; only 8-bit storage and independent channels are studied. PTB-XL and ESC-50 are split by patient and recording, not by official folds. Hirata's reconstruction is our implementation, with a recurrence threshold the method does not prescribe and our own conversion to samples. Sipers' runs on the circular RP it requires, with a frequency threshold set in an exploratory sweep and confirmed on calibration windows outside the evaluated samples. The sampling-rate study covers one dataset, and σ_2/σ_1 is an empirical indicator, not an error bound.

8. Conclusions

We compared, under one protocol, how much of a time series survives its image for widely used image representations and XIRP, on seven datasets from six domains. The signed recurrence plot with a geometric decoder, GASF on $[0, 1]$, GADF on a restricted angular range and XIRP are exactly decodable before quantisation under stated conditions, and after 8-bit storage they keep what classifiers trained on original signals need. The unsigned recurrence plot is recovered only up to a reflection; the binary recurrence plot and

the Markov transition field are many-to-one and lose most of the waveform with the decoders evaluated. For recurrence plots, exact recovery needs a non-degenerate geometry, a non-degenerate overlap and one sign-informative pair, and its stability after storage depends on the conditioning of the states, which in the resampled PTB-XL study worsened as the sampling rate grew. A natural line of research is image generation constrained to valid encodings. For the signed RP, a valid image is a signed Euclidean distance matrix with consistent signs whose adjacent states overlap; for GADF, a matrix on the restricted branch. How to enforce these constraints during generation, or to project a generated image onto them, remains open. Series decoded from such images could then be assessed with the fidelity and utility measures used here.

CRediT authorship contribution statement

Adriano García-Giralda Milena: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Francisco García Moreno:** Conceptualization, Methodology, Supervision, Writing – review & editing. **José Antonio Jiménez Madrid:** Methodology, Formal analysis, Writing – review & editing. **María Bermúdez Edo:** Supervision, Writing – review & editing.

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Data availability

All datasets are public and cited. Code, manifests, configurations and derived results: `https://github.com/adrianoggm/time-series-image-reconstruction`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude Code to support software implementation and debugging, LaTeX formatting, manuscript organisation and language refinement. The authors reviewed and validated all resulting code, procedures and text, and take full responsibility for the methodology, results, interpretations and content of the manuscript.

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